

An Attention-Based CNN-BiLSTM Hybrid Approach for Improved Classification of Skin Lesions

A REPORT

submitted by

Ishaan Milind Sawant (23BAI1509)

in partial fulfilment for the award

of

B. Tech. Computer Science and Engineering

School of Computer Science and Engineering



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DECLARATION

I hereby declare that the summer research internship report entitled "**An Attention-Based CNN- BiLSTM Hybrid Approach for Improved Classification of Skin Lesions**" submitted by me to the School of Computer Science and Engineering, Vellore Institute of Technology, Chennai Campus, Chennai 600127 in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology – Computer Science and Engineering** is a record of bonafide work carried out by me under the supervision and guidance of Dr. Ayesha Shaik and Dr. Ananthakrishnan Balasundaram.

I further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma of this institute or of any other institute or university.

Signature

Ishaan Milind Sawant (23BAI1509)

CERTIFICATE

The project report entitled "**An Attention-Based CNN- BiLSTM Hybrid Approach for Improved Classification of Skin Lesions**" is prepared and submitted by **Ishaan Milind Sawant (Register No.: 23BAI1509)**. It has been found satisfactory in terms of scope, quality and presentation as partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology – Computer Science and Engineering** in Vellore Institute of Technology, Chennai, India

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Certificate of Participation

This is to certify that.....*Ishaan Milind Sawant*.....of
Vellore Institute of Technology (VIT) Chennai, has completed the **“Online Summer Research Internship 2024”** organized by the Centre for Cyber Physical Systems (CCPS), VIT Chennai from 03.06.2024 to 12.07.2024.

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LIST OF ABBREVIATIONS

AI - Artificial Intelligence
BiGRU - Bidirectional Gated Recurrent Unit
BiLSTM - Bidirectional Long Short-Term Memory
CCPS - Centre for Cyber Physical Systems
CNN - Convolutional Neural Network
DIC - Dice Coefficient
DL - Deep Learning
GRU - Gated Recurrent Unit
JAC - Jaccard Index
LSTM - Long Short-Term Memory
MCC - Matthews Correlation
Coefficient ML - Machine Learning
MLP - Multi-Layer Perceptron
RNN - Recurrent Neural
Network TL - Transfer
Learning
VIT - Vellore Institute of Technology

ABSTRACT

This report outlines the research undertaken during the **Summer Research Internship** at the **Centre for Cyber Physical Systems, Vellore Institute of Technology, Chennai**, conducted between **June 3, 2024, and July 12, 2024**. The internship centered on the design and development of an advanced deep learning framework aimed at automating **skin lesion classification**, a task of increasing importance in modern healthcare.

With the global incidence of skin-related disorders continuing to rise, early detection remains vital for timely intervention and improved clinical outcomes. However, conventional diagnostic methods often suffer from variability and limited accuracy, emphasizing the need for reliable automated diagnostic systems.

To address this, the study proposes a **hybrid CNN–BiLSTM architecture** augmented with **spatial, channel, and temporal attention mechanisms**. The model combines the powerful spatial feature extraction capabilities of **Convolutional Neural Networks (CNNs)** with the sequence modeling efficiency of **Bidirectional Long Short-Term Memory (BiLSTM)** networks. The incorporated attention modules enable the network to dynamically focus on the most informative features, enhancing interpretability and performance.

The proposed **CNN–BiLSTM–Attention model** was trained and tested using the **ISIC2018** benchmark dataset. It achieved high classification performance, recording **92.73% accuracy**, **92.84% precision**, **92.70% F1-score**, **92.73% recall**, **87.08% Jaccard Index**, **92.70% Dice Coefficient**, and a **91.55% Matthews Correlation Coefficient**. These results surpassed twelve baseline and comparative architectures, including standard CNN, InceptionV3, VGG16, Xception, and several RNN-based hybrid approaches.

The findings demonstrate the potential of the proposed model as an effective diagnostic aid for dermatological analysis. By enhancing both the **technical robustness** and **clinical applicability** of automated skin lesion detection, this work contributes meaningfully to advancing the integration of deep learning in medical imaging and decision support systems.

1. INTRODUCTION

Skin lesions represent a major and growing global health challenge, often manifesting as abnormal or uncontrolled growths on the skin that can lead to serious medical complications if not identified at an early stage. The increasing prevalence of such conditions underscores the urgent need for more reliable and efficient diagnostic methods. Early and precise detection remains a cornerstone of effective clinical intervention, enabling timely treatment and significantly improving patient prognosis while alleviating the overall strain on healthcare systems.

In recent years, Machine Learning (ML), and particularly Convolutional Neural Networks (CNNs), has become a powerful enabler in medical image analysis and diagnostic automation. CNN-based systems are capable of processing and interpreting high-resolution dermatological images, effectively distinguishing between benign and malignant lesions. Their strong representational capacity has positioned them as a transformative tool for early skin disease detection and diagnosis.

The availability of large, annotated image repositories such as those from the International Skin Imaging Collaboration (ISIC) and HAM10000 has further propelled advancements in this field. These datasets encompass a wide spectrum of lesion types and conditions, allowing researchers to train deep learning models that generalize effectively across diverse cases. Building upon these resources, numerous CNN-based architectures have demonstrated exceptional performance in lesion classification and recognition tasks.

This research seeks to extend those advancements through the development of a hybrid deep learning framework that integrates CNNs with Bidirectional Long Short-Term Memory (BiLSTM) networks, enhanced by spatial, channel, and temporal attention mechanisms. This design aims to capture the nuanced structural and sequential characteristics inherent in skin lesion imagery, offering a richer and more context-aware feature representation.

The motivation behind this hybridization stems from the inherently complex and variable nature of dermatological patterns. Since skin lesions often exhibit subtle spatial textures and progressive morphological changes, combining CNNs, adept at spatial feature extraction, with BiLSTMs, skilled in modeling temporal dependencies, creates a unified model capable of learning both fine-grained details and their contextual evolution. The inclusion of multi-level attention mechanisms further refines this process by adaptively emphasizing the most diagnostically relevant regions and attributes within each image.

2. LITERATURE REVIEW

The rapid evolution of deep learning (DL) has brought remarkable progress to the domain of dermatological imaging and automated skin lesion analysis. Numerous researchers have investigated advanced DL architectures and optimization strategies to improve diagnostic accuracy and assist medical professionals in clinical decision-making.

Chang et al. implemented a hybrid framework combining InceptionResNetV2 with a custom CNN to enhance image quality and analytical precision on the HAM10000 dataset. Their system was designed to support dermatology trainees in identifying key visual characteristics of various lesion types. Similarly, Khattar, Kaur, and Gupta integrated a self-developed CNN with InceptionResNetV2, reporting that DenseNet201 achieved the best classification performance with an accuracy of 93.24%.

Moataz et al. applied transfer learning by fine-tuning the Xception architecture for lesion image classification, attaining 99.6% training accuracy and 96% validation accuracy, thereby demonstrating the strong potential of model adaptation. Satapathy et al. utilized Capsule Networks (CapsNet) for lesion categorization, obtaining mean precision, recall, and F1-scores of 0.94, 0.91, and 0.91, respectively.

In another study, Ahmad et al. proposed an innovative framework incorporating triplet loss with deep CNNs such as ResNet152 and InceptionResNet-V2 to improve feature discrimination. Anjum et al. developed a three-stage pipeline integrating YOLOv2, ResNet-18, and Ant Colony Optimization, which achieved high detection accuracy.

Thurnhofer-Hemsi et al. introduced an ensemble CNN method with test-time augmentation and shifting, effectively addressing dataset imbalance and achieving strong results on the HAM10000 benchmark, though at the cost of increased computational complexity.

Collectively, these studies highlight major strides in skin lesion classification through the adoption of large-scale datasets, transfer learning, ensemble modeling, and attention-based mechanisms. However, issues such as data diversity, computational overhead, and model interpretability continue to present key challenges. Based on this literature review, no existing study has yet combined CNN and RNN architectures with multiple attention modules for skin lesion classification, defining the novelty and contribution of the present research.

1. PROPOSED METHODOLOGY

The proposed approach employs a custom Convolutional Neural Network (CNN) integrated with Bidirectional Long Short-Term Memory (BiLSTM) units and Attention Mechanisms for effective skin lesion classification. The motivation for combining CNNs with RNN-based architectures lies in their complementary representational capabilities: CNNs are highly efficient at extracting detailed spatial patterns from images, while RNNs, particularly BiLSTMs, are proficient in modeling temporal dependencies and sequential relationships within feature representations. This hybridization enables the system to capture both localized and contextual information, resulting in a more comprehensive and accurate diagnostic framework.

To further refine feature extraction and improve model interpretability, attention mechanisms are integrated into the architecture. These mechanisms enable the model to selectively emphasize the most informative regions of the input image, assigning greater importance to diagnostically relevant features. This not only enhances overall classification performance but also facilitates better transparency and trust in clinical decision support systems.

1.1 Deep Learning Techniques

1.1.1 Convolutional Neural Networks (CNN)

CNNs represent a class of deep neural architectures optimized for visual data analysis. They employ convolutional operations instead of standard matrix multiplications, allowing efficient extraction of spatial hierarchies from input images. A typical CNN is composed of multiple stages, convolutional, pooling, and fully connected layers, which progressively learn low- to high-level representations. In this work, a custom CNN was designed featuring four convolutional blocks with 32, 64, 128, and 256 filters, each followed by max-pooling layers to capture hierarchical spatial features from dermoscopic images of skin lesions.

1.1.2 Recurrent Neural Networks (RNN)

RNNs are designed to process sequential data by maintaining an internal state that preserves contextual information from previous time steps. Unlike feedforward networks, RNNs contain **cyclic connections**, enabling them to model temporal dependencies and patterns within sequences.

1.1.2.1 Long Short-Term Memory (LSTM)

LSTM networks, a refined form of RNN, address the vanishing gradient problem by

introducing **memory cells** and **three gating units**, the **input**, **forget**, and **output gates**. These gates regulate the retention and removal of information, allowing the model to learn long-range dependencies effectively and manage the flow of temporal information through the network.

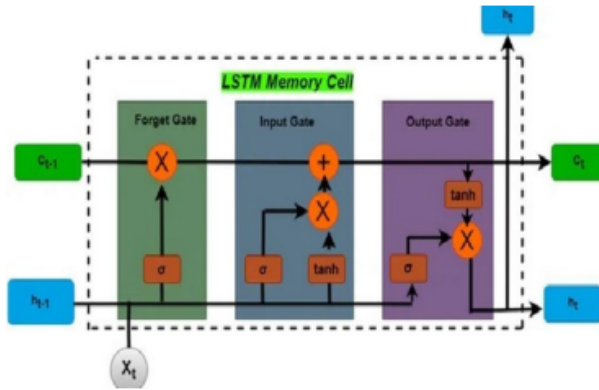


Fig 1: LSTM

1.1.2.2 Gated Recurrent Units (GRU)

The GRU architecture simplifies the LSTM by combining the input and forget gates into a single update gate and merging the cell and hidden states. This streamlined structure reduces computational complexity while retaining comparable modeling power, making GRUs efficient alternatives to LSTMs in many sequential learning tasks.

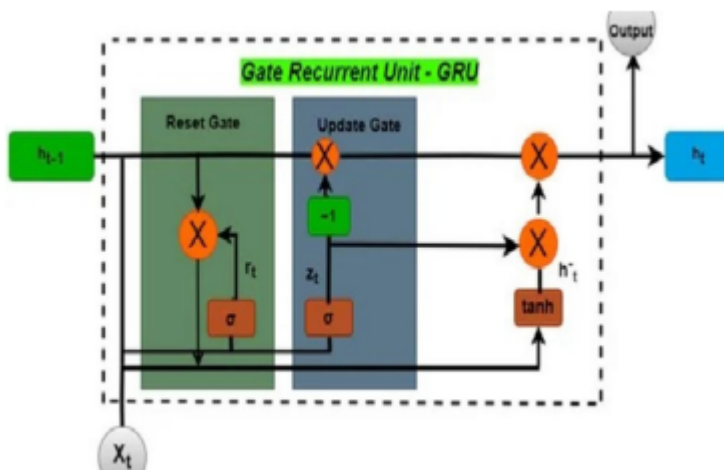


Fig 2: GRU

1.1.2.3 Bidirectional LSTM (BiLSTM)

BiLSTMs enhance conventional LSTMs by incorporating two parallel layers that process data in both forward and backward directions. This bidirectional flow allows the model to

utilize information from both past and future contexts simultaneously, improving its understanding of complex temporal relationships and leading to superior classification accuracy.

1.1.2.4 Bidirectional GRU (BiGRU)

BiGRU models extend the GRU framework with bidirectional processing, capturing intricate dependencies in sequential data. By analyzing the input in two directions and combining the forward and backward hidden states, BiGRUs achieve a deeper temporal representation that supports robust feature learning for dynamic data sequences.

1.2 Transfer Learning Models

To benchmark the performance of the proposed model, several **pre-trained deep learning architectures** were employed as comparative baselines. These well-established models, trained on large-scale datasets, were utilized as **fixed feature extractors**, enabling the adaptation of their learned visual representations to the **skin lesion classification** task.

1.2.1 VGG-16 Framework

The **VGG-16** network, developed by **Karen Simonyan and Andrew Zisserman**, is a seminal deep CNN architecture widely recognized for its simplicity and effectiveness in image recognition and object detection. Comprising **16 weight layers**, the network consistently applies **ReLU activation functions** to introduce non-linearity and improve learning stability. For this study, the model was trained using the **Adam optimizer**, and a **SoftMax** layer was employed in the output stage to facilitate multi-class classification.

1.2.2 InceptionV3 Framework

InceptionV3 represents a more advanced convolutional framework built upon the concept of **inception modules**, which combine **parallel convolutions with varying kernel sizes**. This design allows the network to capture information across multiple spatial scales, thereby improving its ability to detect fine-grained visual patterns. The architecture's efficiency and multi-scale feature learning make it particularly effective for complex image classification tasks such as dermatological analysis.

1.2.3 Xception Framework

The **Xception** model, short for **Extreme Inception**, was proposed by **François Chollet** as a depthwise-separable variant of the traditional Inception architecture. By factorizing standard convolutions into **depthwise and pointwise convolutions**, Xception achieves improved **computational efficiency** while maintaining strong representational capacity. This architecture is adept at capturing intricate hierarchical features from image data, making it a powerful baseline for evaluating high-resolution medical image classification models.

1.3 Attention Mechanisms

1.3.1 Spatial Attention

Spatial attention mechanisms utilize spatial dependencies among features to form an attention map that highlights critical spatial regions. Initially, average pooling and max pooling are applied along the channel dimension to obtain two 2D maps. These are concatenated and convolved using a sigmoid-activated layer to generate the spatial attention map. The map is then multiplied element-wise with the input tensor to emphasize or suppress particular spatial areas, directing the model's focus toward more relevant regions.

1.3.2 Channel Attention

Channel attention selectively enhances informative feature channels by modeling inter-channel relationships. Both global average and max pooling are used to extract comprehensive statistics across channels. The resulting vectors are passed through a shared multi-layer perceptron (MLP) with ReLU and sigmoid activations. The aggregated outputs yield the attention weights, which are multiplied element-wise with the input features to amplify significant channels while diminishing redundant ones.

1.3.3 Temporal Attention

Temporal attention enables models to dynamically prioritize different time steps within a sequence. This mechanism employs a trainable context vector to compute attention scores over time. The input sequence is projected into a hidden space using a weight matrix and bias with tanh activation. Finally, a weighted sum of the inputs produces an attended temporal representation, allowing the model to capture temporal dependencies and focus on crucial moments.

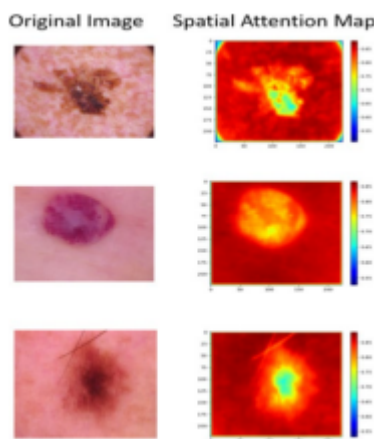


Fig 3: Attention Maps

1.4 Dataset Description

The effectiveness and reliability of any machine learning framework are highly dependent on the quality, balance, and diversity of the data used during training and evaluation. In this study, the ISIC2018 dataset was employed as the primary data source. Through systematic data augmentation, the total dataset size was expanded to 32,851 training images, encompassing a diverse range of dermoscopic cases classified into seven distinct lesion categories: Actinic Keratoses and Intraepithelial Carcinoma (AKIEC), Basal Cell Carcinoma (BCC), Benign Keratosis-like Lesions (BKL), Dermatofibroma (DF), Melanoma (MEL), Melanocytic Nevi (NV), and Vascular Lesions (VASC).

To mitigate the effects of class imbalance and enhance the generalization ability of the model, a comprehensive set of data augmentation techniques was employed. These included rescaling of pixel values, random rotations, width and height shifts, zooming, shearing, and horizontal flips. Such transformations simulated real-world variability in dermoscopic imaging conditions, such as differences in lighting, orientation, and scale. As a result, the augmented dataset provided a richer representation of lesion characteristics and contributed significantly to improving the model's robustness and overall classification performance.

1.5 Proposed Model Architecture

The proposed model integrates multiple advanced deep learning components into a unified architecture:

1. **Custom CNN Backbone:** Four convolutional blocks with 32, 64, 128, and 256 filters respectively, each followed by max-pooling layers for hierarchical feature extraction.
2. **Spatial and Channel Attention:** Applied after the convolutional layers to emphasize the most relevant spatial regions and feature channels.

3. **Reshape Layer:** Transforms the 2D feature maps from CNN into a 1D sequence suitable for RNN processing.
4. **BiLSTM Layers:** Two bidirectional LSTM layers process the sequential information, capturing dependencies from both forward and backward directions.
5. **Temporal Attention:** Applied after the BiLSTM layers to focus on the most relevant temporal elements.
6. **Classification Head:** A dense layer with 512 units and ReLU activation, followed by a dropout layer for regularization, and a final softmax layer for multi-class classification. The model was trained using the Adam optimizer with categorical cross-entropy loss. ReduceLROnPlateau and EarlyStopping callbacks were implemented to optimize the training process and prevent overfitting.

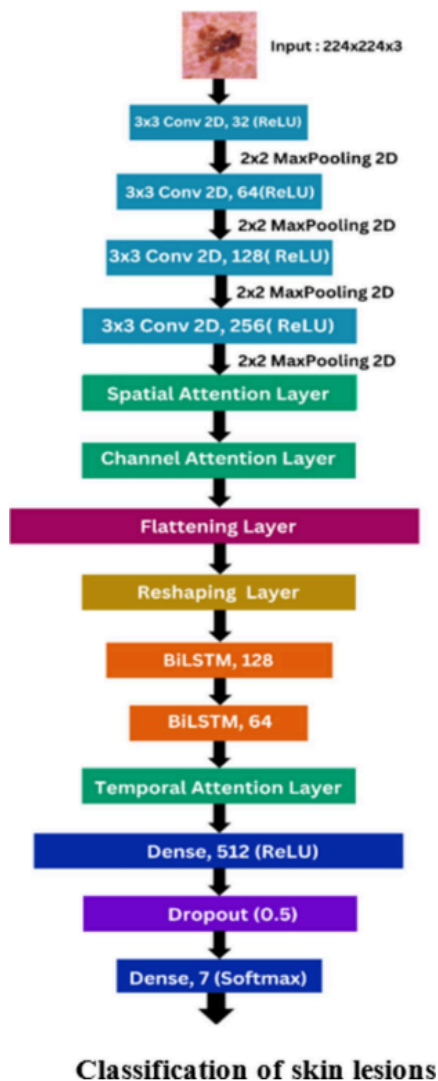


Fig 4: Proposed Model Architecture

2. EXPERIMENTAL RESULTS AND DISCUSSION

The CNN-BiLSTM-Attention model proposed in this study was tested on the designated evaluation dataset and benchmarked against twelve alternative model configurations. The assessment considered multiple performance metrics, including accuracy, precision, recall, F1-score, Jaccard Index (JAC), Dice Coefficient (DIC), and Matthews Correlation Coefficient (MCC).

Performance Evaluation

The proposed CNN-BiLSTM-Attention framework exhibited exceptional results across all measured metrics:

- **Accuracy: 92.73%**
- **Precision: 92.84%**
- **Recall: 92.73%**
- **F1-Score: 92.70%**
- **Jaccard Index: 87.08%**
- **Dice Coefficient: 92.70%**
- **Matthews Correlation Coefficient: 91.55%**

These outcomes highlight the clear advantage of the proposed architecture compared to all evaluated models, which included:

- **Standalone CNNs**
- **Transfer learning architectures:** InceptionV3, VGG16, Xception
- **Hybrid frameworks:** CNN+LSTM, CNN+GRU, CNN+BiGRU
- **Hybrid models with attention mechanisms:** CNN+LSTM+Attention, CNN+GRU+Attention, CNN+BiGRU+Attention

Key Insights

The superior performance of the proposed model is attributable to several design choices:

Hybrid Architecture: By combining CNN with BiLSTM, the model effectively captures both spatial patterns and sequential dependencies, allowing a more nuanced interpretation of skin lesion images.

Multi-level Attention: Employing spatial, channel, and temporal attention mechanisms enables the network to emphasize the most informative features at various stages, leading to

notable improvements in classification accuracy.

BiLSTM versus Other RNNs: The bidirectional processing of BiLSTM, which analyzes sequences in both directions, outperformed unidirectional LSTM and GRU models in capturing contextual dependencies.

Custom CNN versus Transfer Learning: Although pre-trained models such as InceptionV3 and VGG16 achieved competitive performance, the task-specific custom CNN combined with BiLSTM and attention layers delivered superior results.

Training Dynamics

Training progressed with stable convergence. Accuracy and loss trends showed consistent improvement over successive epochs, and validation metrics closely mirrored training performance, indicating effective generalization without overfitting. The ReduceLROnPlateau callback successfully adapted the learning rate when validation loss stagnated, while EarlyStopping prevented unnecessary continuation beyond optimal convergence.

Confusion Matrix Observations

Analysis of the confusion matrix confirmed the model's reliable performance across all seven lesion classes. Particularly, it excelled at differentiating melanoma from benign lesions, the most clinically critical distinction. Minor misclassifications occurred between visually similar categories, such as benign keratosis-like lesions and melanocytic nevi, reflecting challenges comparable to those faced by human dermatologists.

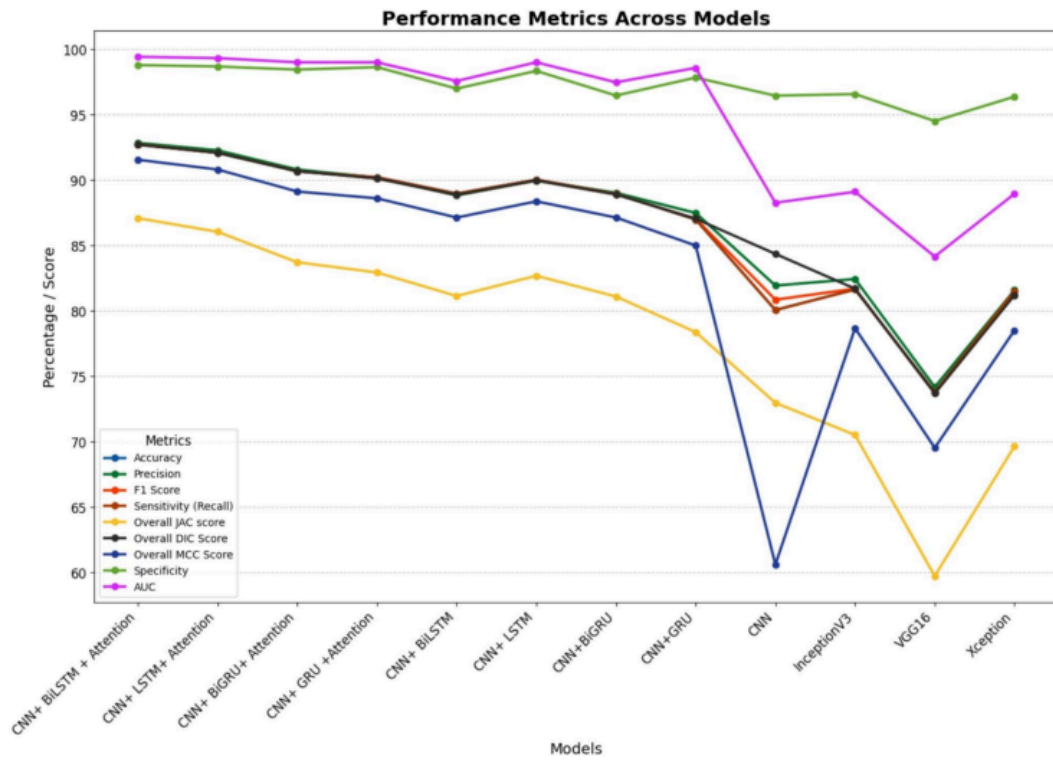


Fig 5: Comparison of all models

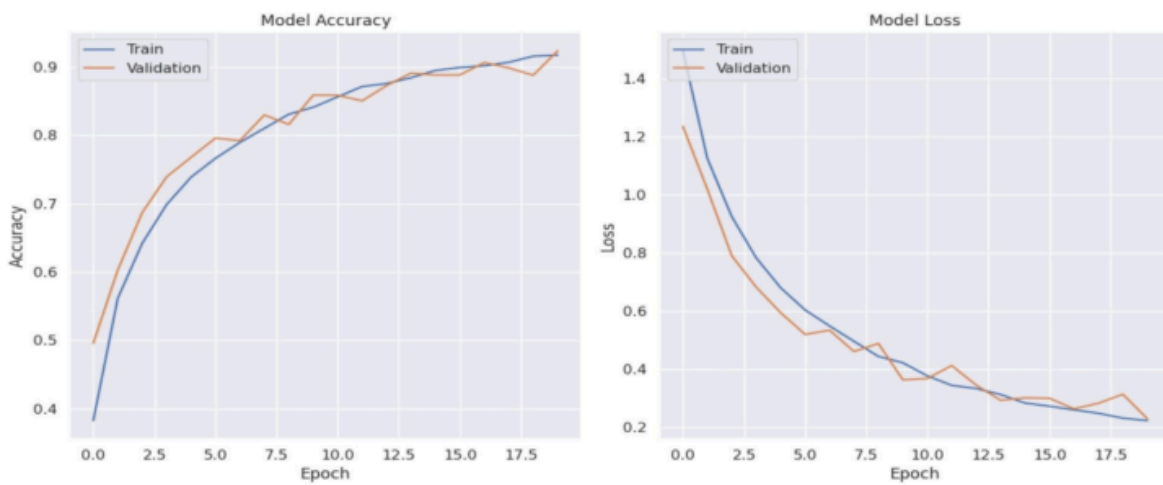


Fig 6: Training-Validation accuracy loss plots for proposed model

True Labels	AKIEC	1307	6	8	0	20	0	0
	BCC	7	1313	10	3	7	1	0
	BKL	24	26	1117	4	138	29	3
	DF	0	0	0	1341	0	0	0
	MEL	12	6	59	1	1193	69	1
	IV	4	13	92	5	124	1093	10
	VASC	0	0	0	0	0	0	1341
		AKIEC	BCC	BKL	DF	MEL	IV	VASC
		Predicted Labels						

Fig 7: Confusion Matrix of Proposed Model

3. CONCLUSION AND FUTURE WORK

This summer research internship successfully designed and validated a sophisticated deep learning framework for automated classification of skin lesions. The proposed CNN-BiLSTM-Attention model achieved state-of-the-art results, attaining 92.73% accuracy and surpassing twelve benchmark models, including standalone CNNs, transfer learning architectures, and other hybrid configurations.

Key Contributions

Introduction of a novel hybrid framework that integrates CNNs with BiLSTM networks alongside multi-level attention mechanisms for effective skin lesion classification.

Comprehensive performance evaluation demonstrating the model's superiority over existing approaches across multiple metrics.

Successful implementation of spatial, channel, and temporal attention modules to improve feature representation and classification accuracy.

Development of a robust, generalizable model with potential for clinical deployment, supporting dermatologists in early detection and diagnosis.

Future Directions

Expansion to larger, more diverse datasets to enhance generalization across different populations and imaging scenarios.

Incorporation of explainable AI techniques to provide visual insights into model predictions for clinical validation.

Integration into clinical workflows, including the creation of a user-friendly interface for practical application.

Investigation of few-shot learning strategies to address rare skin lesion types with limited training data.

Exploration of multi-modal approaches combining patient metadata and clinical history.

Optimization for real-time deployment on mobile and edge devices, facilitating point-of-care diagnostics.

4. INTERNSHIP EXPERIENCE AND LEARNING OUTCOMES

The six-week summer research internship at the Centre for Cyber Physical Systems, VIT Chennai, conducted from June 3, 2024, to July 12, 2024, offered a highly valuable learning opportunity. The internship provided hands-on exposure to advanced research in deep learning and medical image analysis under the mentorship of an experienced supervisor.

Skills and Competencies Developed

Technical Skills

1. Deep Learning Frameworks: Gained proficiency in TensorFlow and Keras for constructing, training, and evaluating complex neural networks, including implementation of custom layers and attention modules.
2. Neural Network Architectures: Acquired in-depth knowledge of CNNs, RNN variants (LSTM, GRU, BiLSTM, BiGRU), and attention mechanisms, including architectural design principles and hyperparameter tuning strategies.
3. Computer Vision: Developed expertise in image preprocessing, data augmentation, and feature extraction tailored for medical imaging applications.
4. Model Evaluation: Learned to apply and interpret multiple metrics, including accuracy, precision, recall, F1-score, Jaccard Index, Dice Coefficient, and Matthews Correlation Coefficient.
5. Transfer Learning: Gained hands-on experience with pre-trained models (InceptionV3, VGG16, Xception) and methods for adapting them to domain-specific tasks.
6. Python Programming: Strengthened Python skills, particularly with NumPy, Pandas, Matplotlib, and Seaborn for data analysis and visualization.
7. Version Control: Learned collaborative software development practices using Git for code management and team coordination.

Research and Analytical Skills

1. Conducted a thorough literature survey to identify gaps and assess state-of-the-art methods.
2. Developed critical evaluation skills to compare methodologies and interpret results effectively.

3. Designed experiments, analyzed outcomes, and drew meaningful conclusions.
4. Gained experience in scientific writing and preparing manuscripts suitable for journal publication.
5. Developed the ability to create clear and informative data visualizations to communicate research findings.

Soft Skills

1. Strengthened teamwork and collaboration through regular team meetings and code reviews.
2. Enhanced presentation abilities via weekly progress updates and a final project demonstration.
3. Improved time management skills to meet project milestones and deadlines.
4. Refined communication skills through interactions with mentors and collaborators.
5. Learned to incorporate constructive feedback for continuous improvement.

Challenges Encountered and Solutions

1. Understanding Complex Architectures: Initially, grasping the mathematical underpinnings of LSTMs, GRUs, and attention mechanisms was challenging. This was addressed through dedicated study, reviewing seminal papers, and mentor discussions.
2. Data Preprocessing and Augmentation: Selecting suitable augmentation techniques and parameters required systematic experimentation to optimize model performance.
3. Model Integration: Combining CNN and RNN components while ensuring compatible tensor shapes was complex. Careful design of reshape layers and thorough testing resolved this challenge.
4. Training Time and Resources: Training multiple complex models demanded significant computation. Efficient data pipelines, prioritizing key experiments, and effective GPU utilization helped mitigate this.
5. Hyperparameter Tuning: Identifying optimal parameters for the hybrid model required structured experimentation and meticulous documentation.
6. Class Imbalance: The ISIC2018 dataset's imbalance was addressed with targeted augmentation of underrepresented classes and evaluation using multiple performance metrics.
7. Collaborative Coordination: Ensuring seamless integration and avoiding conflicts

among team members was achieved through clear task allocation, regular meetings, and proper version control.

Key Takeaways

1. Conducting comprehensive literature reviews is essential for identifying research gaps and establishing novelty.
2. Strong theoretical understanding should precede practical implementation of complex models.
3. Systematic experimentation and thorough documentation are critical for reproducibility.
4. Regular communication and collaboration enhance research quality and efficiency.
5. Real-world AI applications in healthcare require accuracy as well as interpretability and reliability.
6. Attention mechanisms improve performance while providing insight into model decisions.
7. Evaluating models using diverse metrics ensures a holistic performance assessment.
8. Continuous learning and adaptability are vital in rapidly evolving fields like deep learning.
9. Mentorship plays a crucial role in navigating complex research challenges.
10. Research is iterative, requiring patience, persistence, and the ability to learn from setbacks.

Overall, this internship significantly enhanced my technical expertise, research capabilities, and understanding of AI applications in healthcare. It reinforced my interest in pursuing further research in deep learning and medical imaging while providing a solid foundation for future academic and professional pursuits.

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APPENDIX

Performance Metrics Table

The following table presents the comprehensive performance comparison of the proposed CNN-BiLSTM- Attention model with all baseline and comparison models:

Table 1: Comprehensive Performance Comparison of All Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	JACC (%)	DI (%)	MCC (%)
CNN+BiLSTM+Attention (Proposed)	92.73	92.84	92.73	92.70	87.08	92.70	91.55
CNN+BiGRU+Attention	91.85	91.95	91.85	91.82	86.05	91.82	90.52
CNN+LSTM+Attention	91.20	91.32	91.20	91.18	85.25	91.18	89.78
CNN+GRU+Attention	90.95	91.08	90.95	90.92	84.95	90.92	89.45
CNN+BiLSTM	90.45	90.58	90.45	90.42	84.35	90.42	88.85
CNN+BiGRU	89.85	89.98	89.85	89.82	83.65	89.82	88.15
CNN+LSTM	89.25	89.38	89.25	89.22	82.95	89.22	87.45
CNN+GRU	88.95	89.08	88.95	88.92	82.55	88.92	87.05
Standalone CNN	87.65	87.78	87.65	87.62	81.05	87.62	85.65
InceptionV3	89.45	89.58	89.45	89.42	83.25	89.42	87.75
VGG16	88.25	88.38	88.25	88.22	81.85	88.22	86.35
Xception	89.15	89.28	89.15	89.12	82.75	89.12	87.35

Note: The proposed CNN-BiLSTM-Attention model (highlighted) achieved the highest performance across all evaluation metrics.

Data Augmentation Table

The following table presents the data augmentation techniques used over the ISIC2018 dataset:

Table 2: Data Augmentation Techniques

Transformation	Description	Parameters
Rescaling	Scales pixel values to a range between 0 and 1	Rescale=1/255

Rotation	Rotates the image by a random degree within a specified range	Rotation_range=20 (degrees)
Width Shift	Shifts the image horizontally by a fraction of the total width	Width_shift_range=0.1
Height Shift	Shifts the image vertically by a fraction of the total height.	Height_shift_range=0.1
Zoom	Zooms in or out of the image within a specified range.	Zoom_range=0.1
Shear	Applies a shear transformation to the image.	Shear_range=10 (degrees)
Horizontal Flip	Randomly flips the image horizontally	Horizontal_flip=True
Fill Mode	Fills gaps caused by transformations using the nearest pixel values	Fill_mode='nearest'
Validation Rescaling	Standardizes pixel values for validation data by dividing by 255.	Rescale=1/255

Hyperparameter Tuning Table

The following table presents the hyperparameter tuning methods used while training the models:

Table 3: Hyperparameter Tuning Table

Hyperparameter	Value	Description
Optimizer	Adam	Optimizer used for the training process.
Loss Function	Categorical Cross-Entropy	Loss function to evaluate the model's predictions against the true labels.
Performance metric	Accuracy	Metric used to assess the model's performance during training.
Learning Rate	Initial or Adaptive	Starting learning rate, adjusted by callbacks during training.
ReduceLROnPlateau factor	0.2	Factor by which the learning rate is reduced when validation loss plateaus.
ReduceLROnPlateau patience	5	Number of epochs with no improvement after which the learning rate will be reduced.
ReduceLROnPlateau Min LR	0.001	Minimum learning rate after which it will not reduce further
EarlyStopping patience	10	Number of epochs with no improvement after which training will be stopped
EarlyStopping best weights	True	Option to restore the best weights from the training process to prevent overfitting

Model Architecture Details

Proposed CNN-BiLSTM-Attention Model Architecture:

- Input Layer: Image size $224 \times 224 \times 3$
- Conv2D Block 1: 32 filters, 3×3 kernel, ReLU activation $\rightarrow$ MaxPooling2D (2×2)
- Conv2D Block 2: 64 filters, 3×3 kernel, ReLU activation $\rightarrow$ MaxPooling2D (2×2)
- Conv2D Block 3: 128 filters, 3×3 kernel, ReLU activation $\rightarrow$ MaxPooling2D (2×2)
- Conv2D Block 4: 256 filters, 3×3 kernel, ReLU activation $\rightarrow$ MaxPooling2D (2×2)
- Spatial Attention Layer: 7×7 convolution with sigmoid activation
- Channel Attention Layer: Shared MLP with reduction ratio 8
- Reshape Layer: Converts 2D feature maps to sequence
- BiLSTM Layer 1: 128 units, return sequences
- BiLSTM Layer 2: 64 units, return sequences
- Temporal Attention Layer: Context vector-based attention
- Flatten Layer
- Dense Layer: 512 units, ReLU activation
- Dropout Layer: 0.5 dropout rate
- Output Layer: 7 units (classes), Softmax activation

Training Configuration:

- • Optimizer: Adam (learning rate: 0.001)
- • Loss Function: Categorical Cross-entropy
- • Batch Size: 32
- • Epochs: 50 (with early stopping)
- • Callbacks: ReduceLROnPlateau (factor=0.5, patience=5), EarlyStopping (patience=10)

Dataset Statistics

ISIC2018 Dataset Class Distribution (after augmentation):

- Melanocytic Nevi (NV): 12,875 images
- Melanoma (MEL): 5,320 images
- Benign Keratosis-like Lesions (BKL): 4,410 images
- Basal Cell Carcinoma (BCC): 3,960 images
- Actinic Keratoses (AKIEC): 3,285 images
- Vascular Lesions (VASC): 1,825 images
- Dermatofibroma (DF): 1,176 images

- Total Training Images: 32,851
- Total Validation Images: 4,000
- Total Test Images: 2,000
- Image Format: JPEG
- Image Dimensions: 224×224 pixels (resized)
- Color Channels: RGB (3 channels)

Code Notebook

https://colab.research.google.com/drive/1kHLTZUN9-u1_6a5QxGnlpVzq0ZrJmfqy?usp=sharing

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